Homework 5 (Cones)

February 19, 2026

Problems 1, and 2 are worth 10 points each. Problems 3 and 4 are worth 20 points each.

1)

Prove the following:

a)

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Let C be a cone. Then

$$T(C) = \{T(\vec{x}) \,|\vec{x} \in C\}$$

is also a cone.

b)

Let C_1 and C_2 be cones. Then

$$C_1 \times C_2 = \{(\vec{x}, \vec{y}) \,|\vec{x} \in C_1, \vec{y} \in C_2\}$$

is also a cone.

2)

Suppose C is a cone and also a polyhedron. Show that C has at most one vertex.

3)

Let $S \subseteq \mathbb{R}^n$ be a set. Define the dual cone of S as

$$S_* = \{\vec{y} \in \mathbb{R}^n | \vec{x} \cdot \vec{y} \geq 0, \ \forall \vec{x} \in S\}$$

a)

Show that S_* is a cone.

b)

Let $\mathbb{R}^{+n} = \{\vec{x} \in \mathbb{R}^n | \vec{x} \geq 0\}$. Show $\mathbb{R}^{+n}$ is a cone.

c)

Show $\mathbb{R}^{+n}_* = \mathbb{R}^{+n}$. Any cone with this property is said to be self-dual.

d)

The polar cone of a set S is defined as

$$S_o = \{\vec{y} \in \mathbb{R}^n | \vec{x} \cdot \vec{y} \leq 0, \ \forall \vec{x} \in S\}$$

Show $S_o = -S_*$.

4)

Determine if the following sets are cones. If the set is a cone, prove this. If a set is not a cone, show why not.

a)

$$S = \{\vec{x} \in \mathbb{R}^n | \, \|\vec{x}\| = 1\}$$

b)

The set

$$S = \left\{(\vec{x}, r) \in \mathbb{R}^{n+1} | \vec{x} \in \mathbb{R}^n, \text{ and } \|\vec{x}\| \leq r\right\}$$

c)

The set of all positive semi-definite matrices. Recall a matrix is positive semi-definite if $\vec{x}^T A\vec{x} \geq 0$ for all $\vec{x} \in \mathbb{R}^n$. In notation:

$$S = \left\{A \in \mathbb{R}^{n \times n} | \vec{x}^T A\vec{x} \geq 0 \; \forall \vec{x} \in \mathbb{R}^n\right\}$$

d)

$$S = \left\{A \in \mathbb{R}^{n \times n} | \det(A) \geq 1\right\}$$